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Fundamentals of Financial Mathematics
- Home Assignment -

Design of Structured Products:
JPMorgan Chase & Co

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November 2026

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1 Market Data and General Setting

In this report, we design two structured products on a single underlying, the stock of JPMorgan Chase & Co (JPM), which is listed on the New York Stock Exchange and traded in U.S. dollars (USD). The first product is a Partially Principal Protected Note (PPPN), the second an Airbag Note. For each product, we describe the payoff, replicate it with a zero-coupon bond and European options using real market prices, and estimate its expected performance with a Monte Carlo simulation. All computations and figures can be reproduced with the accompanying Python notebook.

1.1 Market Data

All market data was collected from Yahoo Finance[1] on November 12, 2025, at 12:00 PM Eastern Time, during regular U.S. trading hours. We take this moment as $t = 0$. The snapshot contains the spot price of JPM, $S_0 = 320.41$, and the full option chain for the maturity date of January 15, 2027. All prices and payoffs in this report are expressed in USD.

Risk-Free Rate The time from November 12, 2025, to January 15, 2027, is 14 months, so $T = \frac{14}{12} \approx 1.167$. For the risk-free rate r, we take the 1-year and 2-year U.S. Treasury par yields on November 12, 2025 (3.65% and 3.56%) and interpolate linearly to 14 months, which gives $r = 3.635\%$. We assume this rate to be constant and continuously compounded throughout the report.

Option Data On November 12, 2025, the market quoted 47 European calls and 41 European puts on JPM, all maturing on January 15, 2027, with strikes between 105 and 470. Some strikes are not quoted for puts, which explains the smaller number of puts. Fig. 1 shows the quoted prices over the strike range.

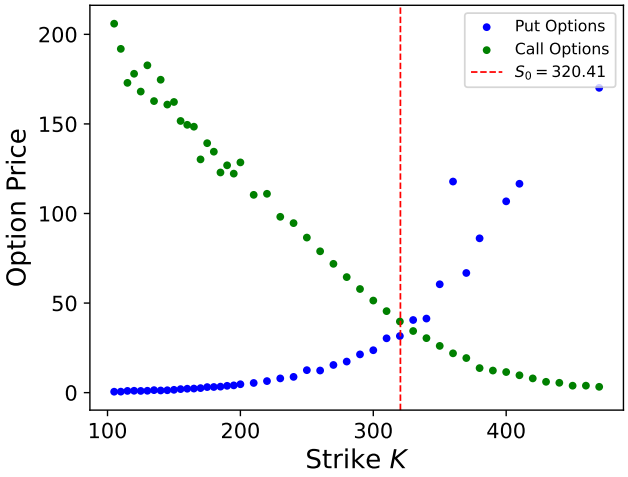


Figure 1: **Option Prices.** Market prices of the European puts and calls on JPM with maturity January 15, 2027, quoted on November 12, 2025.

[1] `https://finance.yahoo.com/quote/JPM/`

2 Partially Principal Protected Note

The PPPN balances protecting the investment against taking part in the movements of the stock. At maturity on January 15, 2027, the investor is guaranteed to get back 90% of the invested capital. On top of that, the note pays a premium if JPM ends up outside a corridor around today's price S_0 (see Fig. 3). The premium grows with the distance from the corridor, so large moves are rewarded in either direction.

JPM is a good fit for such a product. Over the past year, the stock rose from \$239.56 on November 12, 2024, to \$320.41 at $t = 0$, an increase of about 34%. Moves of this size are exactly what the premium pays for, while the investor never risks more than 10% of the capital.

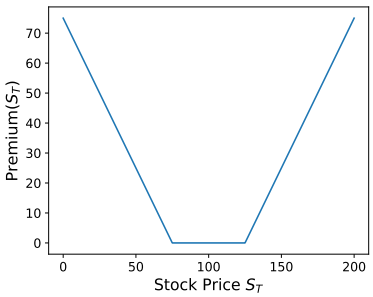
2.1 Payoff

The payoff of the PPPN has two parts: a protected base and a premium that depends on the stock price S_T at maturity.

Principal Protection To secure 90% of the invested amount N, part of the capital is invested in a zero-coupon bond. Its present value is $0.9Ne^{-rT}$, and it grows to $0.9N$ at maturity, whatever happens to the stock.

Premium Two strikes $K_1 < S_0 < K_2$ define a corridor in which no premium is paid. Outside the corridor, the premium is proportional to how far the stock has moved:

$$\text{Premium}(S_T) = \begin{cases} \frac{N}{S_0}(K_1 - S_T), & \text{if } S_T < K_1, \\ 0, & \text{if } K_1 \le S_T \le K_2, \\ \frac{N}{S_0}(S_T - K_2), & \text{if } S_T > K_2. \end{cases}$$



The plot on the right shows an example with $S_0 = 100$, $K_1 = 75$ and $K_2 = 125$. For the rest of this section, we set $N = S_0$, so that $\frac{N}{S_0} = 1$.

2.2 Replication

The bond provides the capital protection. The premium is replicated by buying $\frac{N}{S_0}$ European puts with strike K_1 and $\frac{N}{S_0}$ European calls with strike K_2, both with maturity T.

Replication Cost The issuer also charges a margin M to cover its profit and hedging costs. We express it as a fraction m of the investment, $M = mN$. At $t = 0$, the bond, the options and the margin must add up to the amount invested:

$$N = 0.9Ne^{-rT} + \frac{N}{S_0}\big[EP(K_1, T) + EC(K_2, T)\big] + mN. \tag{1}$$

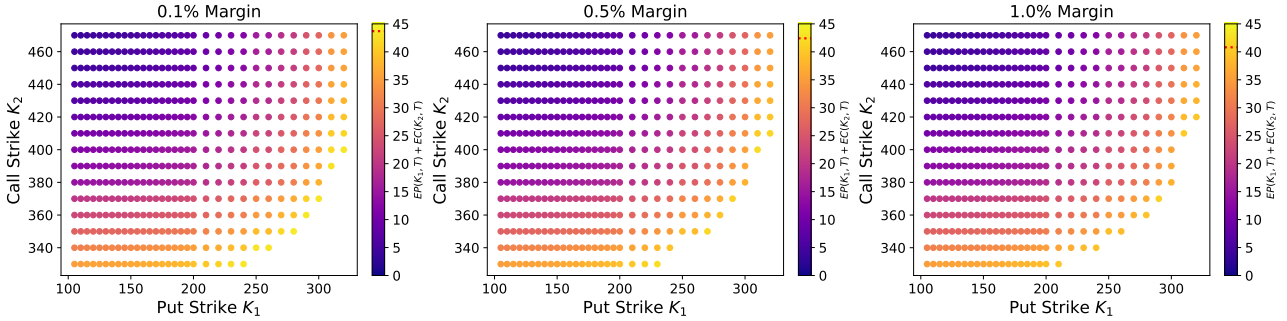


Figure 2: **Valid Strike Pairs.** Strike pairs (K_1, K_2) whose combined option cost fits within the budget (2) for margins of 0.1%, 0.5% and 1.0%. The colour gives the total cost of both options, and the dotted red line in each colour bar marks the budget.

Choosing K_1 and K_2 Rearranging (1) gives the budget available for the two options:

$$EP(K_1, T) + EC(K_2, T) = S_0 \left(1 - 0.9e^{-rT} - m\right). \tag{2}$$

Strikes are only quoted in steps of 5 or 10, so this equality can usually not be met exactly. We therefore only require the left-hand side to be at most the budget; any amount left over goes to the issuer. Fig. 2 shows all strike pairs (K_1, K_2) that satisfy this condition for margins of 0.1%, 0.5% and 1.0%. Options with strikes close to S_0 are more expensive, because a narrow corridor makes a premium more likely. A higher margin leaves less money for the options and therefore forces a wider corridor.

We choose a margin of $m = 0.5\%$, which leaves the issuer a reasonable profit while keeping the product attractive. With $S_0 = 320.41$, $r = 3.635\%$ and $T = 1.167$, the option budget is 42.41. For the investor, the best strikes are those with the narrowest corridor $K_2 - K_1$. The narrowest valid corridor has a width of 80, and among the pairs that reach it we take the one that uses most of the budget:

$$EP(270, T) + EC(350, T) = 15.45 + 26.09 = \mathbf{41.54} \leq \mathbf{42.41}.$$

The note therefore uses $K_1 = 270$ and $K_2 = 350$. Fig. 3 places this corridor on the price history of JPM.

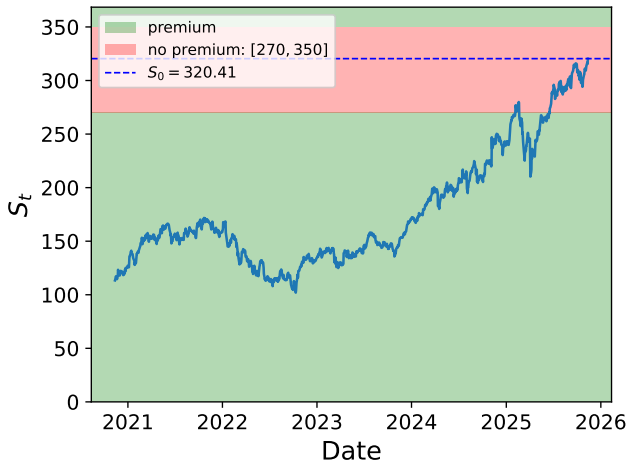


Figure 3: **PPPN Corridor for JPM.** Closing prices of JPM over the past five years together with the premium zones of the note. A premium is paid below $K_1 = 270$ and above $K_2 = 350$ (green), but not in between (red).

2.3 Simulation

To estimate how the note might perform, we simulate JPM as a geometric Brownian motion, discretised with the Euler–Maruyama scheme over the 294 trading days until maturity. The drift $\mu = 23.47\%$ is the average daily log-return of JPM over the past five years, annualised. The volatility $\sigma = 28.9\%$ is the implied volatility of the options around the chosen strikes[2]. The options are priced under the risk-neutral measure, but the simulation uses the real-world drift because we want to know how the product is likely to perform. Further details are given in Appendix A.

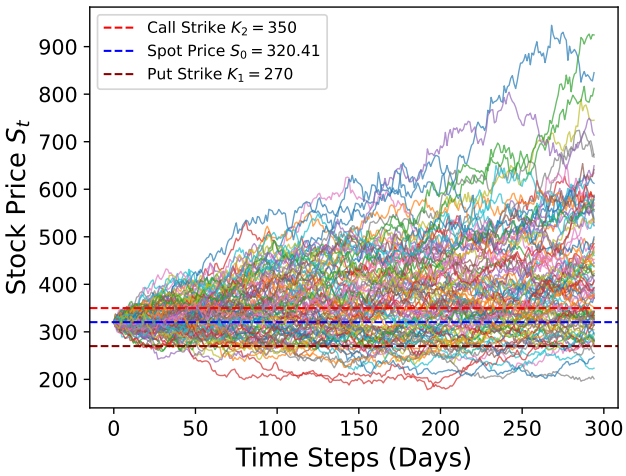


Figure 4: **Simulation.** 100 of the 10^4 simulated paths of JPM over 294 trading days. The horizontal lines mark the strikes K_1 and K_2 and the spot price S_0.

Fig. 4 shows a sample of the paths. Over 10^4 paths, the stock leaves the corridor in 76.9% of the cases, and the expected premium at maturity and its present value are

$$\hat{\mathbb{E}}[\text{Premium}(S_T)] = 95.57, \qquad e^{-rT}\,\hat{\mathbb{E}}[\text{Premium}(S_T)] = 91.60 > 41.54 = EP(K_1, T) + EC(K_2, T).$$

The gap does not mean the options are cheap. It comes from the historical drift of JPM, which is far above the risk-free rate. The expected total payoff is

$$0.9N + \hat{\mathbb{E}}[\text{Premium}(S_T)] = 288.37 + 95.57 = 383.94,$$

which corresponds to a growth factor of $\frac{383.94}{S_0} \approx 1.198$, or about 20% on the initial investment.

[2] `https://optioncharts.io/options/JPM/volatility-skew?option_type=all&expiration_dates=2027-01-15:m&strike_range=270,350`

3 Airbag Note

The Airbag Note is designed for investors who want to profit from further growth of JPM and still be protected against moderate losses. At maturity on January 15, 2027, the payoff depends on the final price of JPM:

- If the stock ends at or above its initial price S_0, the investor receives the invested capital plus a share p, the participation rate, of the stock's gain.
- If the stock ends below S_0 but above the airbag level A, the investor still receives the full capital.
- Only if the stock falls below the airbag level does the payoff drop, in proportion to the stock's performance.

The airbag works as a cushion against moderate declines, which makes the note suitable for risk-conscious investors who still want to take part in JPM's upward trend.

3.1 Payoff

The payoff at maturity depends on S_T, the participation rate $p \in (0, 1)$ and the airbag level $A \in (0, S_0)$. It is convenient to write it as $N \cdot \varphi_{p,A}(S_T)$, where the payoff factor $\varphi_{p,A}$ scales the initial investment N:

$$\text{Payoff}_{p,A}(S_T) = N \cdot \varphi_{p,A}(S_T) = N \cdot \begin{cases} 1 + \frac{p}{S_0}(S_T - S_0), & \text{if } S_0 \le S_T, \\ 1, & \text{if } A \le S_T < S_0, \\ \frac{S_T}{A}, & \text{if } S_T < A. \end{cases} \tag{3}$$

The three cases correspond to a gain ($\varphi > 1$), preserved capital ($\varphi = 1$) and a loss ($\varphi < 1$). Working with φ lets us analyse the product independently of the amount invested. For this product, we set the airbag level to $A = 0.8\,S_0 = 256.33$. The payoff is shown in Fig. 5.

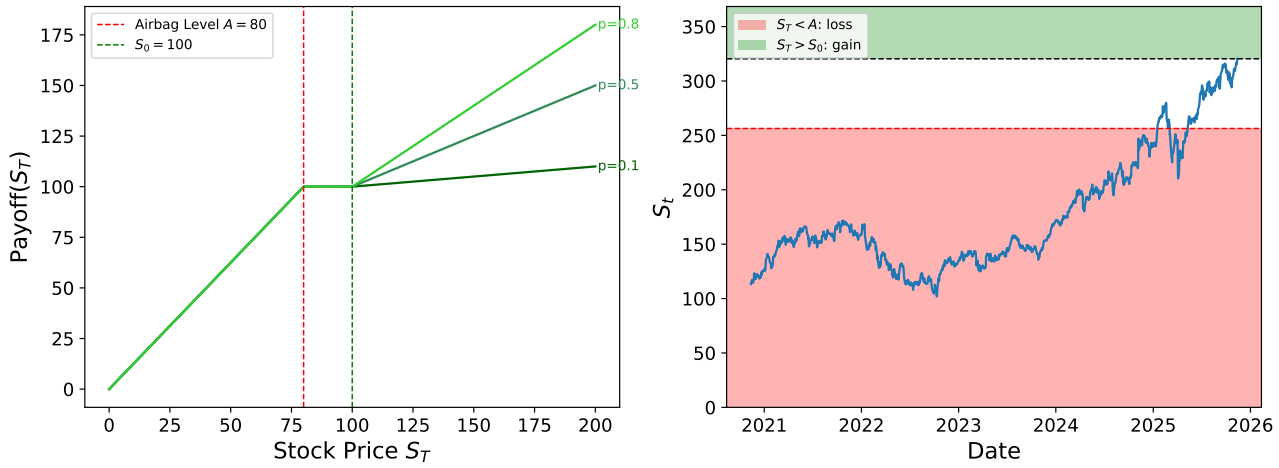


Figure 5: **Airbag Payoff.** Left: payoff for $N = 100$, $S_0 = 100$ and $A = 80$ with different participation rates. Right: closing prices of JPM over the past five years with the airbag level $A = 256.33$ and the spot price $S_0 = 320.41$. The investor gains above S_0 (green), keeps the capital between A and S_0 (white) and makes a loss below A (red).

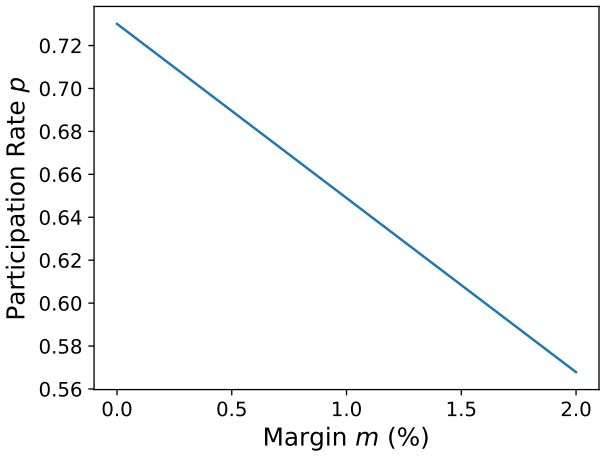


Figure 6: **Participation vs. Margin.** Participation rate $p(m)$ for margins m between 0% and 2%.

Parameter	Value
r	0.03635 (3.635%)
T	1.167 (14 months)
S_0	320.41
A	256.33
$EC(S_0, T)$	39.49
$EP(A, T)$	12.42
α	≈ 0.730
β	≈ -8.11

Table 1: **Parameters.** Inputs for the participation rate $p(m)$, where α is the intercept and β the slope.

3.2 Replication

The payoff (3) can be rewritten as

$$\text{Payoff}_{p,A}(S_T) = N\left(1 + \frac{p}{S_0}(S_T - S_0)^+ - \frac{1}{A}(A - S_T)^+\right).$$

This form shows the replicating portfolio directly: a zero-coupon bond that pays N at maturity, a long position in $\frac{Np}{S_0}$ European calls with strike S_0, and a short position in $\frac{N}{A}$ European puts with strike A. Including the issuer's margin $M = mN$ as before, the cost at $t = 0$ must equal the investment:

$$\begin{aligned} N &= N\left(e^{-rT} + \frac{p}{S_0}EC(S_0, T) - \frac{1}{A}EP(A, T) + m\right) \\ \Rightarrow \quad 1 &= e^{-rT} + \frac{p}{S_0}EC(S_0, T) - \frac{1}{A}EP(A, T) + m. \end{aligned}$$

Since both strikes are fixed, the only free parameters are the margin m and the participation rate p. Solving for p gives a linear function of the margin:

$$p(m) = \underbrace{\left(1 - e^{-rT} + \frac{1}{A}EP(A, T)\right)\frac{S_0}{EC(S_0, T)}}_{\alpha} + m\underbrace{\left(-\frac{S_0}{EC(S_0, T)}\right)}_{\beta}. \tag{4}$$

Neither $S_0 = 320.41$ nor $A = 256.33$ is a quoted strike. We therefore interpolate the option prices linearly between the two nearest strikes: $EC(320.41, T)$ between the calls at 320 and 330, and $EP(256.33, T)$ between the puts at 250 and 260. The resulting values are listed in Table 1.

Fig. 6 shows this linear relationship for margins between 0% and 2%. Without a margin, the investor would participate with $p(0) = 0.730$; at $m = 2\%$, the participation falls to $p(0.02) = 0.568$. As for the PPPN, we choose $m = 0.5\%$, which gives a participation rate of $p(0.005) \approx 0.690$.

3.3 Simulation

We assess the Airbag Note with the same simulated paths as the PPPN (Sec. 2.3), this time evaluating the payoff factor $\varphi_{p,A}(S_T)$. Over 10^4 paths, the note ends with a gain in 76.4% of the cases, preserves

the capital in 16.2%, and makes a loss in only 7.4%. The expected payoff factor is

$$\hat{\mathbb{E}}[\varphi_{p,A}(S_T)] = 1.234 > 1,$$

which means an expected growth of 23.4% on the initial investment. Fig. 7 shows 100 of the simulated paths together with the airbag level.

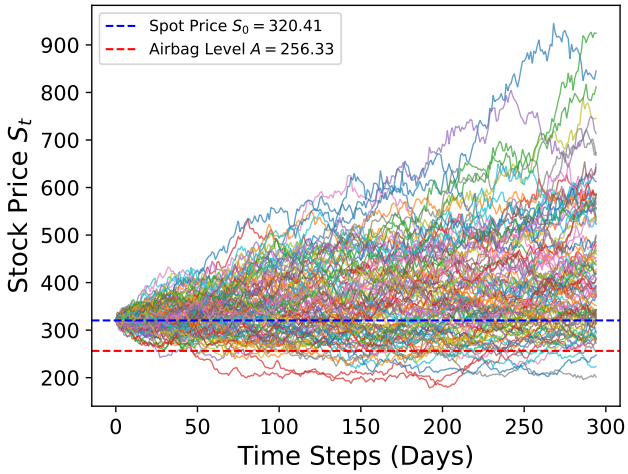


Figure 7: **Simulation.** 100 of the 10^4 simulated paths of JPM. The horizontal lines mark the airbag level A and the spot price S_0.

4 Summary and Conclusion

In this report, we presented two structured products on JPM, a Partially Principal Protected Note (PPPN) and an Airbag Note, both built from a zero-coupon bond and vanilla European options. Table 2 compares them.

Table 2: Comparison of the two structured products.

Aspect	**PPPN**	**Airbag Note**
Structure	90% capital protection. Pays a premium if the stock moves out of the corridor $[270, 350]$ in either direction.	Full capital protection down to the airbag level $A = 256.33$. Participates partially ($p = 0.690$) in gains.
Replication	Zero-coupon bond worth $0.9\,Ne^{-rT}$, long $EP(270, T)$ and long $EC(350, T)$.	Zero-coupon bond worth Ne^{-rT}, long $EC(S_0, T)$ and short $EP(A, T)$.
Issuer margin	0.5%	0.5%
Expected growth (Monte Carlo)	$\hat{\mathbb{E}}[\text{Payoff}(S_T)]/N = 1.198$	$\hat{\mathbb{E}}[\varphi_{p,A}(S_T)] = 1.234$

The PPPN combines partial capital protection with exposure to large moves in either direction. The Airbag Note instead focuses on downside protection: the investor keeps the full capital as long as JPM stays above the airbag level and participates in its gains, while moderate losses are absorbed. Both products are therefore aimed at risk-conscious rather than purely risk-seeking investors, that is, at investors who want their capital protected to some degree but still want to benefit from rising prices.

Monte Carlo simulations based on the historical drift of JPM point to favourable expected performance for both products. For the PPPN, the present value of the expected premium is more than twice the cost of the two options. For the Airbag Note, the expected payoff factor is above one, and a loss occurs in fewer than one in ten simulated scenarios. These estimates rely on JPM's strong performance over the past five years continuing, so they should be read as indicative rather than as a forecast.

A Monte Carlo Simulation

This appendix describes the Monte Carlo simulation used to estimate the expected payoffs of both products. The code is available in the accompanying notebook.

A.1 Model and Discretisation

The stock price follows a geometric Brownian motion with drift μ and volatility σ,

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t,$$

which we discretise with the Euler–Maruyama scheme:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t}\, \epsilon_t,$$

where Δt is the time step and the $\epsilon_t \sim \mathcal{N}(0,1)$ are independent.

A.2 Procedure

1. Simulate n paths of the stock price from S_0 up to maturity T.
2. For each path i, evaluate the quantity of interest X_i at maturity: the premium $(K_1 - S_T)^+ + (S_T - K_2)^+$ for the PPPN, and the payoff factor $\varphi_{p,A}(S_T)$ from (3) for the Airbag Note.
3. Estimate the expectation by the sample mean

$$\mathbb{E}[X] \approx \hat{\mathbb{E}}[X] = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

With $n = 10^4$, the standard error is 1.12 for the expected PPPN premium and 0.003 for the expected Airbag payoff factor.

A.3 Parameters

- Initial stock price: $S_0 = 320.41$
- Maturity: $T = 14/12 \approx 1.167$ years
- Number of time steps: $n_D = 294$, the NYSE trading days between November 12, 2025, and January 15, 2027
- Time step: $\Delta t = T/n_D$
- Drift: $\mu = 252 \cdot \overline{\log(S_{d+1}/S_d)} = 0.2347$, the annualised mean daily log-return of the dividend-adjusted closing price over the past five years
- Implied volatility: $\sigma = 0.289$
- Number of paths: $n = 10{,}000$